



AMD DEVELOPER HACKATHON · ACT II · TRACK 3 – UNICORN

RecipeReel

Turn any cooking video into a perfect, structured recipe.

AMD Instinct MI300X

ROCm

Gemma

Fireworks AI

Whisper · Qwen2.5-VL

01 THE PROBLEM

You found the recipe. It's trapped in a 12-minute video.

🔍 You scrub back and forth

Pausing every few seconds to catch the next step, losing your place, burning the onions.

⌚ The amounts flash for two seconds

“2 cups” appears as on-screen text, then it's gone. Was it 2 or 2½? Cups or ounces?

🗑️ None of it is usable

You can't search it, scale it, shop it, or save it. Hundreds of millions of cooking videos — and not one is machine-readable.

RecipeReel watches the video and **writes the recipe.**

Paste a link — YouTube, TikTok, a Reel — or upload a file. Get back one clean, structured recipe.

One-Pan Garlic Butter Shrimp Pasta

PREP 5m COOK 15m TOTAL 20m SERVES 4 confidence 0.86

INGREDIENTS · QUANTITIES FROM ON-SCREEN TEXT

12 oz linguine
1 lb large shrimp · peeled
4 tbsp butter
5 clove garlic · minced
½ cup dry white wine
½ cup parmesan
— salt & pepper · to taste

STEPS · EACH DEEP-LINKS TO THE EXACT SECOND

00:12 Boil salted water; cook linguine to al dente, reserve pasta water.
00:36 Melt butter over medium heat; add garlic until fragrant.
00:48 Add shrimp; ~2 min per side until pink.
01:02 Deglaze with white wine + lemon; reduce one minute.
01:26 Toss pasta in; finish with parmesan, parsley, chili flakes.

The amounts are never spoken. They're **on the screen.**

Transcript-only tools guess quantities and get them wrong. RecipeReel fuses three signals — and never invents a number.

SPEECH →

The steps

Narration gives the actions, technique, and ordering. Whisper transcribes it with timestamps.

ON-SCREEN →

The quantities

The authoritative amounts live in overlay cards. A vision model reads them verbatim.

TIME →

The deep links

Every step is pinned to the exact second, so a tap jumps you into the video.

Heavy perception on the GPU. Structured reasoning with **Gemma**.

INGEST

yt-dlp + ffmpeg

Fetch video & metadata; extract audio and scene keyframes with exact timestamps.

LISTEN

Whisper-large-v3

Speech → timestamped transcript.

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SEE

Qwen2.5-VL

Frames → captions + on-screen text (the quantities).

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REASON

Gemma 4

Fuse everything → one schema-constrained Recipe (temp 0).

`json_schema`

SERVE

FastAPI

Validate, persist, stream progress.
REST + live SSE.

Every stage is a pluggable provider with CPU / mock fallback – the pipeline degrades, it never crashes.

Real work on real silicon.

The MI300X's 192 GB holds Whisper-large-v3 *and* Qwen2.5-VL resident at once — the throughput-bound perception that justifies the hardware. Gemma does the reasoning, hosted on AMD via vLLM to compete for **Best AMD-Hosted Gemma**.

192_{GB}

HBM3, both models resident

2

GPU workloads: ASR + VLM

1

codebase, CPU→MI300X

06 ROCm masquerades as CUDA

The same `device="cuda"` code runs on a laptop, then unchanged on the MI300X. Perception on-GPU; Gemma on Fireworks or self-hosted on AMD. Nothing is faked.

A production-shaped backend

that runs in one command.

Async job engine

Queue → worker → live SSE progress. SQLite persistence for jobs and recipes.

Clean REST API

Submit URL/upload, poll, stream, fetch recipe, export `schema.org` JSON-LD. Swagger at `/docs`.

Containerized

Slim Dockerfile + an MI300X ROCm image. MIT-licensed, public repo, README.

Pluggable providers

Whisper / Qwen-VL / Gemma swap by env var. Missing key or GPU → graceful fallback.

Offline demo mode

`MOCK_MODE=true` runs the whole pipeline with zero keys — the demo never hard-fails.

Tested & linted

Pytest suite green, ruff-clean, verified end-to-end in Docker.

07 WHY IT'S A COMPANY

Every cooking video is a product waiting to be structured.

◆ Home cooks

Save, scale, and shop any video recipe. The consumer wedge.

◆ Creators & publishers

Auto-generate recipe cards + schema.org rich results — instant SEO and reach from content they already made.

◆ Grocery & commerce

Structured ingredients become a shoppable cart. The recipe is the funnel.

◆ Platform & data

An API that makes video searchable, plus the largest structured cooking dataset as a moat.

The recipe-app and meal-planning market is measured in billions — and it's still built on people retyping videos by hand.



From reel to **recipe.**

One paste. Real quantities, timestamped steps, and structured data — powered by AMD.

github.com/avinmaster/recipe-reel

FastAPI · Python 3.12

MI300X · ROCm · Gemma

RecipeReel — AMD Developer Hackathon: Act II · Track 3 (Unicorn)